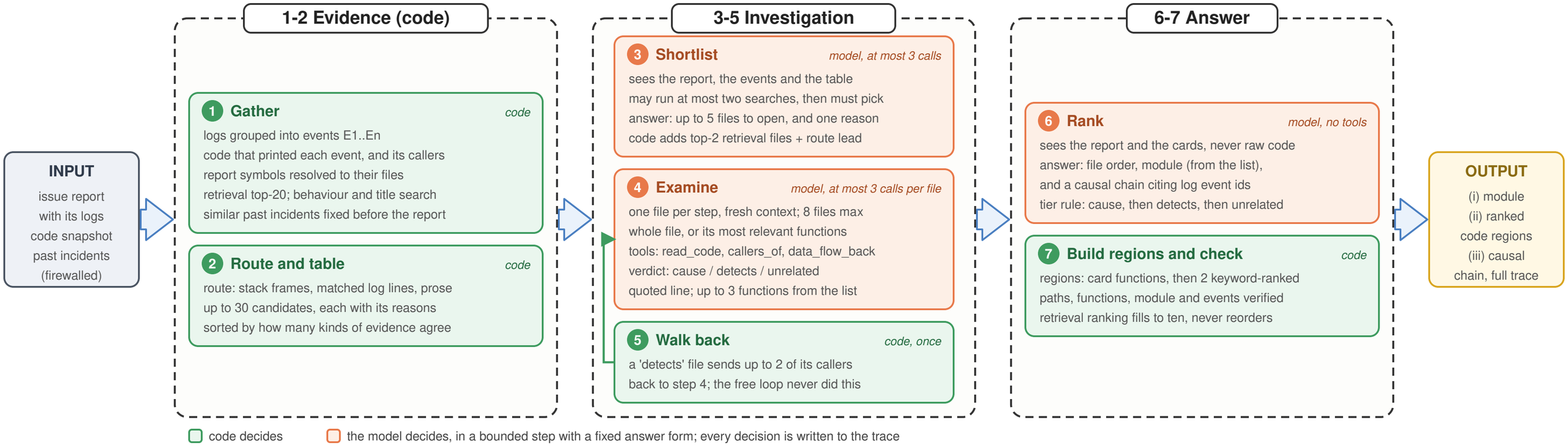
INPUT
issue report
with its logs
code snapshot
past incidents
(firewalled)
1-2 Evidence (code)
1 Gather
code
logs grouped into events E1..En
code that printed each event, and its callers
report symbols resolved to their files
retrieval top-20; behaviour and title search
similar past incidents fixed before the report
2 Route and table
code
route: stack frames, matched log lines, prose
up to 30 candidates, each with its reasons
sorted by how many kinds of evidence agree
3-5 Investigation
3 Shortlist
model, at most 3 calls
sees the report, the events and the table
may run at most two searches, then must pick
answer: up to 5 files to open, and one reason
code adds top-2 retrieval files + route lead
4 Examine
model, at most 3 calls per file
one file per step, fresh context; 8 files max
whole file, or its most relevant functions
tools: read_code, callers_of, data_flow_back
verdict: cause / detects / unrelated
quoted line; up to 3 functions from the list
5 Walk back
code, once
a 'detects' file sends up to 2 of its callers
back to step 4; the free loop never did this
6-7 Answer
6 Rank
model, no tools
sees the report and the cards, never raw code
answer: file order, module (from the list),
and a causal chain citing log event ids
tier rule: cause, then detects, then unrelated
7 Build regions and check
code
regions: card functions, then 2 keyword-ranked
paths, functions, module and events verified
retrieval ranking fills to ten, never reorders
OUTPUT
(i) module
(ii) ranked
code regions
(iii) causal
chain, full trace
code decides
the model decides, in a bounded step with a fixed answer form; every decision is written to the trace